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A Poisson shot noise model for micro-penetration of snow

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ABSTRACT

As a common starting point for the interpretation of SnowMicroPenetrometer (SMP) data the penetration force is interpreted as a superposition of spatially uncorrelated ruptures of structural elements which follow an ideal elastic–brittle response. We re-state this idea and describe the fluctuating penetration force as a Poisson shot noise process. This allows us to derive simple analytical expressions for the cumulants and the covariance of the penetration force in terms of the micromechanical, force-displacement parameters of individual elements. Vice versa, the micromechanical parameters can be estimated from the statistics of the penetration force. We test our method with simulated shot noise processes and real snow profiles and reveal potential limitations of the underlying assumptions. Our model unifies different previous approaches to snow classification which are based on snow penetration resistance.

1. Introduction

The microstructure and stratigraphy of a snow cover is of great importance for various physical processes, such as the radiation balance at the snow surface (e.g. Kuipers-Munneke et al., 2009), heat transfer (e.g. Dadic et al., 2008), failure initiation (e.g. Heierli and Zaiser, 2008) or avalanche formation in general (e.g. Schweizer et al., 2003). All thermal, mechanical and optical properties of snow are influenced by its microstructure. Traditional manual snow cover measurements insufficiently characterize the complex structure of snow. These measurements mainly provide information on crystal type and size, density and a rough measure of hardness (e.g. hand hardness index or 'rammsonde' measurements). Furthermore, manual observations are time consuming and prone to subjectiveness (Painter et al., 2007). By using traditional observations it is therefore not possible to accurately characterize snow microstructure and its physical properties in an unambiguous manner.

In order to obtain more objective data, new measurement techniques were developed to determine structural and physical properties of snow samples with far greater detail. Most of these measurement techniques, such as contact spectroscopy (Painter et al., 2007), near-infrared photography (Matzl and Schneebeli, 2006) or micro-computer tomography (Coléou et al., 1999; Schneebeli and Sokratov, 2004) allow to characterize the snow microstructure with high spatial resolution and provide new insight into physical mechanisms (e.g. Pinzer and Schneebeli, 2009). Portable devices to efficiently conduct in-situ measurements are particularly desired since such devices could potentially replace manual snowpit measurements. Recently, Arnaud et al. (2011) developed a field device to measure optical characteristics of snow from infrared reflectance measurements at cm resolution. Schneebeli and Johnson (1998) developed the SnowMicroPenetrometer (SMP) to obtain mechanical characteristics of the snow microstructure from cone penetration tests (CPT) (Johnson and Schneebeli, 1999; Schneebeli et al., 1999). While CPT are widely used to characterize stratified media in various disciplines of geotechnical engineering (e.g. Huang et al., 2004), the SMP is specifically tailored to probe the mechanical failure of snow in the penetration direction at high spatial resolution (4 μm) by using a small cone (5 mm diameter).

The interpretation of SMP data still lacks a common conceptual starting point. Besides phenomenological indices derived for specific applications (e.g. Bellaire and Schweizer, 2011; Pielmeier and Marshall, 2009; van Herwijnen et al., 2009) there are basically two strategies to derive microstructural snow properties from the SMP force signal: i) a micromechanical approach to characterize the elastic–brittle failure characteristics of individual structural elements (Johnson and Schneebeli, 1999; Marshall and Johnson, 2009 and ii) a purely statistical approach which characterizes snow in terms of low-order moments of the SMP force signal (Satyawali and Schneebeli, 2010; Satyawali et al., 2009). The first approach is based on peak counting and subsequent signal inversion by Monte-Carlo simulations (Marshall and Johnson, 2009), referred to here as the M&J method. The applicability is thereby limited by the spatial resolution of the SMP and cannot be applied to structural element sizes below 0.59 mm (Marshall and Johnson, 2009). In contrast, the statistical approach can be applied to arbitrary force signals, though the interpretation lacks any connection to an underlying physical model.

In this paper we show that both approaches are intimately related and that the microstructural parameters can be deduced from the analytical solution of a stochastic model which underlies the micromechanical

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approach of Johnson and Schneebeli (1999). To this end we show that a slight generalization of the currently accepted model implies a penetration force which can be interpreted as a Poisson shot noise process. This well-known stochastic process is amenable to an analytical treatment which allows us to estimate the micromechanical model parameters by matching theoretical expression and experimental estimates for the mean, the variance and the covariance (or likewise the semivariogram) of the force signal. Within this model our procedure provides a theoretical explanation why the statistical descriptors used in Satyawali and Schneebeli (2010) are useful measures for a snow classification scheme and how they are related to the micromechanical parameters used in Marshall and Johnson (2009). Though our proposed procedure is *a priori* applicable to arbitrary sizes of structural elements we show that the accuracy of parameter estimation is indeed influenced by the spatial resolution of the SMP signal. Finally we apply our method to a real snow profile containing various layers and discuss the potential of our work to unify different previous approaches to develop a snow classification scheme which is based on snow penetration resistance.

2. Stochastic micromechanical model

2.1. Poisson shot noise process representation of the SMP force

We start from the micromechanical model introduced by Johnson and Schneebeli (1999) which is based on the assumption that the SMP force signal is a superposition of spatially uncorrelated, elastic deflections and brittle failures of individual snow microstructural elements. In the present study we mainly focus on the mathematical implications of these assumptions.

Three basic parameters are commonly derived from the penetration resistance measurements (Fig. 1): the rupture strength f, the deflection at rupture δ and the element size L which is the average distance between neighboring elements. In order to relate this simple one dimensional force characteristic to the three dimensional situation below the SMP cone, additional assumptions for the spatial distribution of elements are required. A penetrometer with (projected) cone area A, moving vertically over a distance H penetrates a volume $V = AH$. If the volume contains structural elements of mean size L the average number of elements in the volume is $\overline{N} = AH/L^3$. Johnson and Schneebeli (1999) assume an almost regular assembly of structural elements as shown schematically in the *2d* vertical section on the left in Fig. 2.

Within each vertical column, the sequence of elements is strictly periodic in L. The randomness emerges from the assumption that in each column the distance of the topmost element to the penetrometer is a random variable which is distributed uniformly in $[0, L]$. In the following we drop the assumption of regular arrays and assume that elements are distributed randomly in the volume V as schematically shown on the right hand side in Fig. 2. More precisely, we assume that the positions of the elements are drawn from a homogeneous Poisson point process in *3d* space (Mecke and Stoyan, 2000) which rigorously implements the loose idea of placing N elements "completely at random" in a volume V. Note that the random positioning of elements does not account for the steric exclusion of neighboring elements, i.e. elements can overlap. The point process is uniquely characterized by its number density or intensity λ_{3d} which is related to the expected number of elements $\overline{N}$ in the volume V by $\overline{N} = \lambda_{3d}V = \lambda_{3d}AH$.

It is further assumed in (Johnson and Schneebeli, 1999) that any element in V contributes to the response solely by its deflection in z direction. The force–deflection characteristic of a single element follows an ideal elastic–brittle behavior which can be cast into

$$F(z) = \begin{cases} \frac{f}{\delta} z & 0 < z < \delta \\ 0, & \text{otherwise} \end{cases}. \tag{1}$$

Therefore the superposition signal, i.e. the total force $F_T(z)$ recorded by the SMP, depends solely on the z coordinates of the elements in V. We can therefore restrict ourselves to a one dimensional point process with intensity $\lambda := \overline{N}/H = \lambda_{3d}A$.

Following Johnson and Schneebeli (1999) we define the element size L as the mean distance between two neighboring elements in the volume via $\overline{N}L^3 = AH$. Thus the mean distance L between neighboring elements in *3d* space and the intensity of the *1d* point process are related by

$$\lambda = \frac{A}{L^3}. \tag{2}$$

The total SMP force is the superposition of mechanical responses of all elements in the volume $V = AH$ and can be written as

$$F_T(z) = \sum_{n=1}^{N} F(z - z_n), \tag{3}$$

where the single element contributions $F(z)$ are characterized by (1) and the random positions z_n are drawn from a *1d* Poisson point process with intensity λ which is related to the average distance L between elements via Eq. (2).

The formal representation (3) reveals that our stochastic model of the SMP signal can be regarded as a so called shot noise process (e.g. Papoulis, 1984). Shot noise is usually employed to describe the fluctuations in a time series due to the random arrivals of signal carriers such as the emission of photons in a laser source. Due to the absence of spatial correlations, many statistical properties can be calculated analytically.

2.2. Distributions of rupture strengths

We make an additional generalization of the model by allowing the rupture strength f to be a random variable itself. We assume that f is statistically independent of the Poisson process and characterized by a probability density $p_s(f)$. Averages over the strength distribution are denoted by $\langle \bullet \rangle = \int df\, (\bullet)\, p_s(f)$.

We consider two different cases of strength distributions. First we consider the deterministic case of uniform rupture strengths which can be recovered by a Dirac distribution

$$p_s^{(D)}(f) = \delta(f - f_0). \tag{4}$$

Second, we consider the more realistic case of fluctuating rupture strengths. From a practical point of view any distribution restricted to

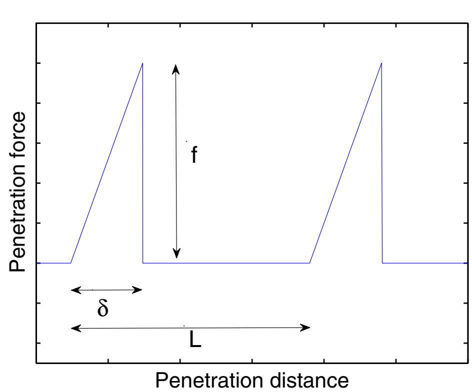

Fig. 1. Schematic representation of the three basic structural parameters in SMP signals: structural element size L, rupture force f and deflection at rupture δ.

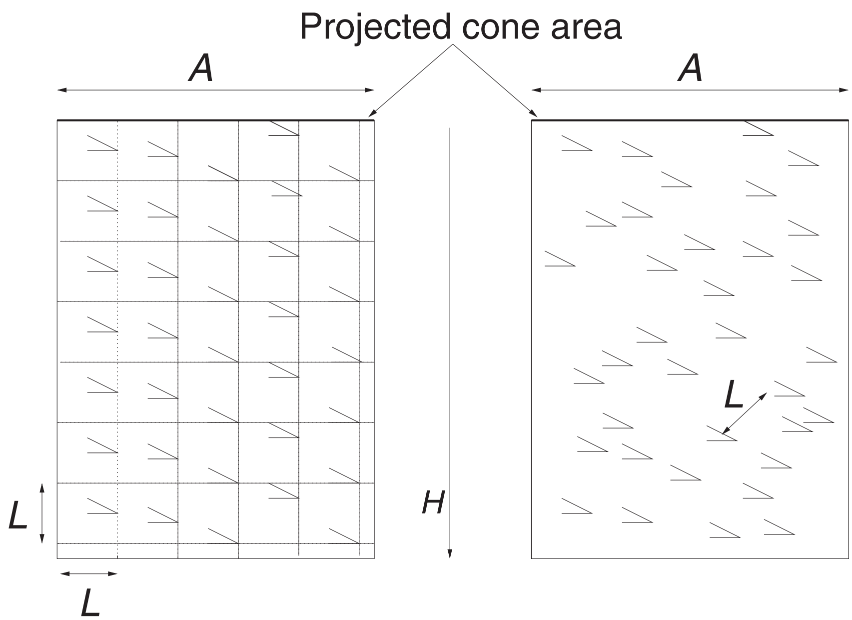

Fig. 2. Schematic of the spatial arrangement of elements as used by (Johnson and Schneebeli, 1999) (Left) or as implied by a Poisson point process (Right).

positive strength values can be used. An obvious choice would be the Weibull distribution, as it is widely used for similar failure phenomena (Kun et al., 2006). However, the influence of the particular choice of the strength distribution becomes irrelevant for large intensities λ, i.e. for large cone areas A or for small L (high snow densities), see Eq.(2). This follows from the central limit theorem which guarantees that the distribution of $F_T(z)$ tends to a Gaussian, provided that the first and second moment $\langle f\rangle$, $\langle f^2\rangle$ of the rupture strength distribution exist (Papoulis, 1984). As a practical implication, the central limit theorem indicates some degree of universality where details of the probability distribution become irrelevant. This justifies a certain freedom to replace the Weibull form by a distribution which is more convenient from a computational point of view. To this end we employ a Gamma distribution

$$p_s^{(G)}(f) = \frac{1}{f\Gamma(\alpha)}(\alpha f/f_0)^{\alpha} exp(-\alpha f/f_0). \tag{5}$$

where the parameter f_0 is the mean rupture force and $1/\alpha$ is the coefficient of variation which determines the shape of the distribution. The uniform case (4) can be recovered in the limit $\alpha\to\infty$ as e.g. shown by Patriarca et al. (2004). The technical advantage of a Gamma distribution over a Weibull distribution will become clear later.

3. SMP force statistics

3.1. Analytical solution for one and two-point functions

In the following we derive simple analytical expressions for statistical characteristics of the force signal in terms of the micromechanical parameters. We always assume a homogeneous point process characterized by a spatially uniform λ. Spatial inhomogeneities of a real SMP signal can be accounted for by applying the homogeneous picture to a moving window. This is equivalent to the moving window approach in Satyawali et al. (2009).

We refer to van Etten (2005) and relate the cumulants κ_n of the shot noise process $F_T(z)$ to the shape function $F(z)$ by

$$\kappa_n = \lambda\left\langle \int_{-\infty}^{\infty} dz\ [F(z)]^n \right\rangle. \tag{6}$$

This is a general result and independent of the shape of the individual (integrable) force–deflection response $F(z)$. It can thus be applied to different failure models. The expression (6) still involves the average over the strength distribution as denoted by angular brackets. Inserting the ideal elastic–brittle failure (1) into (6) and evaluating the integral yields an expression for the cumulants of the SMP force $F_T(z)$ in terms of the moments $\langle f^n\rangle$ of the rupture strength

$$\kappa_n = \frac{\lambda\,\delta\,\langle f^n\rangle}{n+1}. \tag{7}$$

Note that the lowest order cumulants are simply the ensemble mean $\kappa_1 = \left\langle\overline{F_T(z)}\right\rangle$ and the variance $\kappa_2 = \left\langle\overline{(F_T(z)-\kappa_1)^2}\right\rangle$ of the SMP force. The expression (7) reveals that the cumulants of any order, and thus the entire one-point statistics of the penetration force $F_T(z)$ solely depend on the product $\lambda\delta$. Hence, estimating both parameters individually requires higher order correlation functions. To this end we consider the force covariance which is defined by

$$C(r) := \left\langle\overline{(F_T(z)-\kappa_1)(F_T(z+r)-\kappa_1)}\right\rangle. \tag{8}$$

Again we can employ the general result (e.g. Papoulis, 1984) to obtain

$$C(r) = \lambda\left\langle \int_{-\infty}^{\infty} dz\ F(z)F(z+r)\right\rangle. \tag{9}$$

The remaining integral can be computed to yield the exact expression

$$C(r) = \frac{\lambda\delta\left\langle f^2\right\rangle}{3}\left(1-\frac{3}{2}\left(\frac{|r|}{\delta}\right)+\frac{1}{2}\left(\frac{|r|}{\delta}\right)^3\right)\theta(\delta-|r|). \tag{10}$$

Here $\theta(r)$ denotes the Heaviside step function, defined by $\theta(r)=1$ for $r>0$ and $\theta(r)=0$ otherwise. We note that the semi-variogram $G(r)$ of the penetration force which is sometimes used in applications (Satyawali and Schneebeli, 2010) can be immediately obtained from the covariance via $G(r)=\kappa_2-C(r)$. Thus the shot noise model predicts a semivariogram which is of spherical type.

3.2. Analytical parameter estimation by method of moments

The micromechanical parameters can be estimated from Eq. (7) which provides a nonlinear overdetermined system of arbitrary order between measurable one-point statistics (the cumulants of the SMP force) and the micromechanical parameters which characterize the individual force deflection curves. If k micromechanical parameters have to be determined, the first $n=1, 2, \ldots k$ equations of the system (7) can be used to employ the method of moments. Since the system (7) only involves the product $\lambda\delta$ an additional equation is required to estimate the intensity λ of the shot noise process which requires two-point statistics of the SMP force. The simplest estimate for δ can be inferred from the slope of the correlation function (10) at the origin

$$\delta=-\frac{3}{2}\frac{C(0)}{C'(0)}. \tag{11}$$

This procedure of parameter estimation is demonstrated explicitly in the following for both cases of rupture strength distributions.

3.2.1. Uniform rupture strengths

If we consider the deterministic case of uniform rupture strengths given by Eq. (4), the model involves only three parameters, namely the intensity λ of the Poisson process and the two micromechanical parameters δ, f_0. We thus use the first two equations $n=1, 2$ of the system (7), insert $\langle f^n\rangle=f_0^n$ and solve the system analytically, yielding

$$\begin{aligned}\lambda\delta &=\frac{4}{3}\frac{\kappa_1^2}{\kappa_2}\\ f_0 &=\frac{3}{2}\frac{\kappa_2}{\kappa_1}.\end{aligned} \tag{12}$$

Subsequently δ can be eliminated by means of Eq. (11).

3.2.2. Fluctuating rupture strengths

Next we consider the case of Gamma-distributed rupture strengths given by Eq. (5). In this case the model involves four parameters, namely λ and three micromechanical parameters δ, f_0, α. The moments of a Gamma distributed f are given by

$$\langle f^n\rangle=f_0^n\prod_{k=0}^{n-1}\frac{\alpha+k}{\alpha}. \tag{13}$$

Inserting Eq. (13) for $n=1, 2, 3$ into Eq. (7) yields a system of equations with polynomial non-linearities. This system can again be solved analytically, e.g. by using the symbolic algebra software MAPLE. The solution reads

$$\begin{aligned}\lambda\delta &=\frac{6\kappa_1^2\kappa_2}{9\kappa_2^2-4\kappa_3\kappa_1}\\ f_0 &=\frac{9\kappa_2^2-4\kappa_3\kappa_1}{3\kappa_1\kappa_2}\\ \alpha &=\frac{2\left(4\kappa_3\kappa_1-9\kappa_2^2\right)}{9\kappa_2^2-8\kappa_3\kappa_1}.\end{aligned} \tag{14}$$

Subsequently δ can be inferred by means of Eq. (11). Note that for a Weibull distributed f the respective moments $\langle f^n\rangle$ lead to transcendental non-linearities which cannot be handled analytically anymore. This reveals the convenience of our choice to use a Gamma distributed f.

3.3. Practical recipe of SMP parameter estimation

Eqs. (12), (14) provide our main results which open a simple route for an unbiased parameter estimation:

1. Compute the estimate of the the non-central moments of the SMP force in a finite windows with M sample points from the empirical estimator

$$\hat{\mu}_n=\frac{1}{M}\sum_{m=1}^{M}\left[F_T(z_m)\right]^n \tag{15}$$

2. An estimate of the cumulants can be obtained from their relation to the non-central moments

$$\hat{\kappa}_n=\hat{\mu}_n-\sum_{n'=1}^{n-1}\binom{n-1}{n'-1}\hat{\kappa}_{n'}\hat{\mu}_{n-n'} \tag{16}$$

together with $\hat{\kappa}_1=\hat{\mu}_1$. The other two equations required here are explicitly given by $\hat{\kappa}_2=\hat{\mu}_2-\hat{\mu}_1^2$ and $\hat{\kappa}_3=\hat{\mu}_3-3\hat{\mu}_2\hat{\mu}_1+2\hat{\mu}_1^3$

3. An estimate of $\lambda\delta$ and f_0 in the case of uniform strengths or $\lambda\delta$, f_0 and α in the case varying rupture strengths can be obtained from the cumulants via Eqs. (12), (14)
4. The final parameter δ is computed from an unbiased estimate of the covariance according via Eq. (11) using the finite difference approximation $C'(0)=(C_1-C_0)/\varepsilon$. Here C_m, $m=1\ldots M-1$ denotes the discrete covariance and $\varepsilon=4\,\mu\text{m}$ the SMP resolution.

3.4. Comparison to parameter estimation by peak counting

The M&J method for parameter inversion based on peak counting cannot be applied in dense systems since many simultaneous deflections and ruptures mutually mask each other (Marshall and Johnson, 2009). The number of such missed events depends on the sampling frequency, in the SMP case given by $f_s=250$/mm corresponding to a spatial resolution $\varepsilon=1/f_s=0.004$ mm. From the properties of the Poisson point process we derive a theoretical upper limit of the intensity λ below which the method of rupture counting is expected to work properly. A peak, or likewise a rupture, can only be detected if neighboring elements are separated by at least a distance 2ε. The probability P_{all} that all elements are detected is thus equal to the probability that there is at most 1 element in an arbitrarily chosen interval of length $\Delta=4\varepsilon$. Since the expected number of elements in any interval of length Δ is Poisson distributed with mean $\lambda\Delta$ the probability is given by $P_{all}(\Delta,\lambda)=exp(-\lambda\Delta)(1+\lambda\Delta)$. The expected number of detected elements per unit length n_{det} is obtained by multiplication with the expected number of elements per unit length λ, viz $n_{det}(\Delta,\lambda)=P_{all}(\Delta,\lambda)\lambda$. In the theoretical limit of infinite sampling frequency the spatial resolution $\varepsilon=0$ is approached where the number of detected elements equals the true number of elements per mm $n_{det}(0, A/L^3)=A/L^3$. For the finite SMP resolution $\varepsilon=4\,\mu\text{m}$ we obtain $n_{det}(16\,\mu\text{m}, A/L^3)$ as the number of detected elements. Both curves for the number of detected elements per mm are shown in Fig. 3 as a function of element size. For finite spatial resolution the number of detected elements attains a maximum at some element size and decreases rapidly below it. In this regime almost every element is masked by a neighbor and individual elements cannot be discerned anymore. As a consequence, the intensity λ, or likewise the element size L, cannot be estimated anymore by peak counting. Some algebra yields a unique location of the maximum of $n_{det}(\Delta, A/L^3)$ at $L_{min}=\left(2\Delta A/\left(1+\sqrt{5}\right)\right)^{1/3}$ which implies $L_{min}\approx 0.58$ mm for the SMP resolution $\Delta=4\varepsilon=0.016mm$. This compares very well to the value $L_{min}=0.59$ mm which was numerically obtained by Marshall and Johnson (2009) as the lower limit below which signal inversion by peak counting is no longer possible. Our Fig. 3 corresponds to Fig. 4

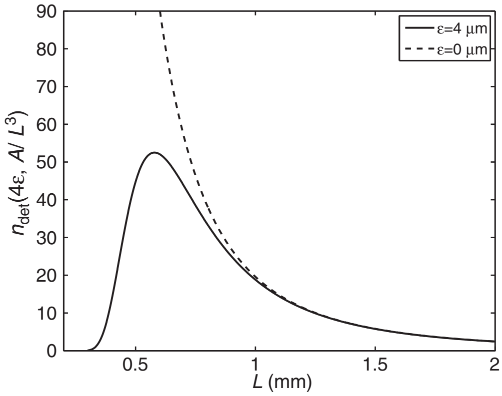

Fig. 3. Comparison of the expected number of detected elements from peak counting for the limiting case of spatial resolution $\varepsilon=0$ (dashed curve) and for the actual SMP resolution $\varepsilon=4\,\mu m$ (solid line).

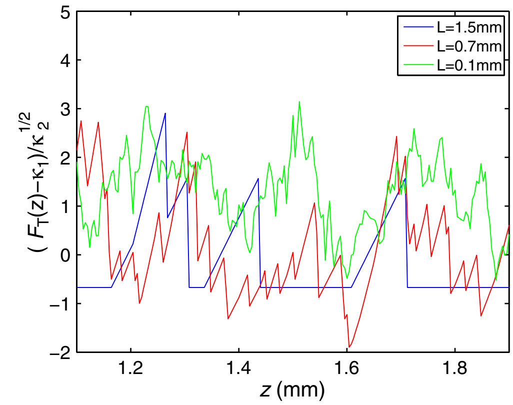

Fig. 4. Overview of normalized simulated force signals obtained for different intensities λ with uniform rupture strengths.

in (Marshall and Johnson, 2009). In contrast, the signal inversion introduced in the previous section can *a priori* be applied to all values of parameters. This will be demonstrated below for element sizes down to $L=0.1$ mm.

4. Simulations of model SMP signals

4.1. Simulations of shot noise processes

To provide some confidence for our analytical solution and test our method of parameter estimation we first employ simulations of shot noise processes as idealized SMP signals. If not stated otherwise we use the true spatial resolution $\varepsilon=4\,\mu m$ of the SMP and the effective cone area $A=19.6\,mm^2$ is fixed such that element size and the intensity of the point process can always be related by $\lambda=A/L^3$.

We simulated a single realization of the shot noise process (3) on a finite spatial domain of size $H=50$ mm for various parameter triples. The number of elements N in each realization was drawn from a Poisson distribution with mean λH and the positions of the elements are independent, identically distributed random variables on $[0, H]$. For each element the strength f_0 is subsequently drawn from the strength distributions (4),(5). For uniform rupture strengths we conducted 128 simulations for all combinations of parameters taken from $f_0=0.1, 0.2, 0.3, 0.4$N, $\delta=0.1, 0.2, 0.3, 0.4$ mm and $\lambda=6, 9, 15, 27, 57, 157, 726, 19600\,mm^{-1}$ ($L\approx 0.1, 0.3, \ldots 1.5$ mm). For fluctuating rupture strengths the same values are used together with $\alpha=10$.

A qualitative overview of the signal forms for uniform rupture strengths and different values of L is given in Fig. 4. While for high $L=1.5$ mm individual peaks of the ideal elastic–brittle response are clearly visible, a low value of $L=0.1$ mm gives rise to an apparent continuous signal. For comparability the signals have been normalized by their mean and variance. The signals for fluctuating rupture strengths are very similar.

The correctness of Eq. (7) is revealed by Fig. 5 for both probability distributions, $p_s^{(D)}$ from Eq. (4) in the case of uniform rupture strengths and $p_s^{(G)}$ from Eq. (5) in the case of a Gamma distributed rupture strengths.

In addition we simulated 100 realization of a shot noise process for $\lambda=157\,mm^{-1}$, $\delta=0.1$ mm, $f_0=0.2$ N on spatial domains of length $H=10$ mm for different spatial resolutions $\varepsilon=1, 4, 16\,\mu m$. The resolution of the SMP corresponds to $\varepsilon=4\,\mu m$. The results of the covariance averaged over all realizations in the case of uniform rupture strengths are plotted in Fig. 6. The exact result Eq. (10) is reasonably well approached at higher spatial resolution. The same quality of convergence is attained for fluctuating rupture strengths.

4.2. Parameter estimation for shot noise processes

In a second step we test the recipe of parameter estimation introduced in the previous section on the simulated shot noise signals. In practice parameters are commonly estimated by applying moving window to the SMP force signal. We therefore subdivided the 50 mm signals into non-overlapping intervals of size $w=1, 5, 10$ mm. For uniform rupture strengths the parameter estimates are shown in Fig. 7. These results reveal the statistical limitations of simulated signals when analyzed in small windows and suggests that e.g. the intensity λ of the shot noise process is generally overestimated, in

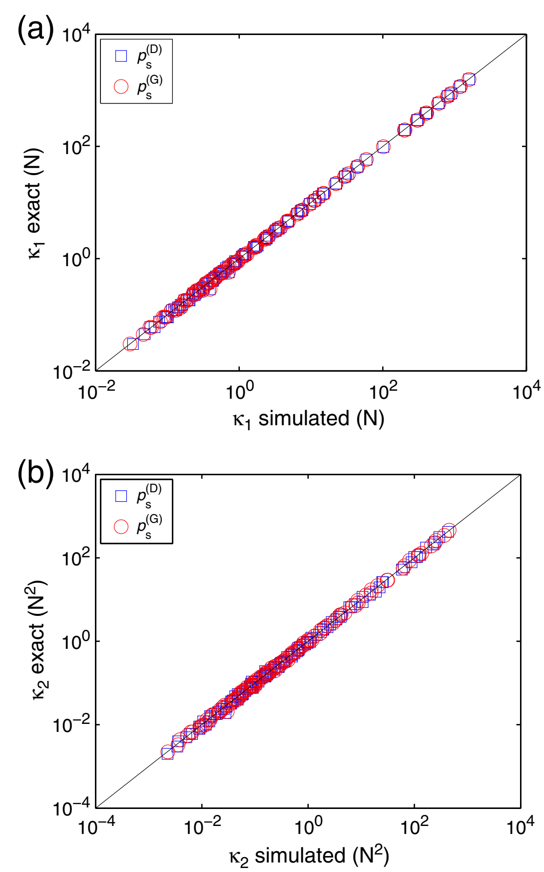

Fig. 5. Comparison of the cumulants obtained from simulations with the exact value from Eq. (7). (a) Mean force ($n=1$). (b) Variance ($n=2$). All parameter combinations for uniform and fluctuating rupture are shown.

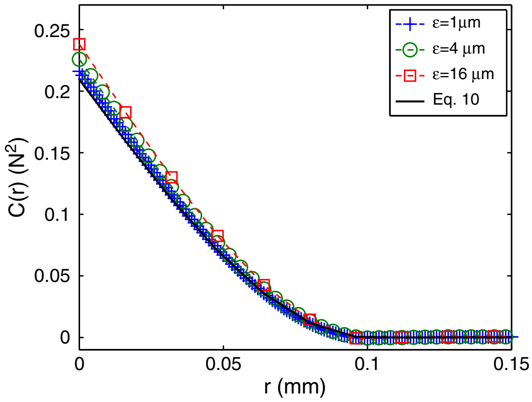

Fig. 6. Comparison of simulated and theoretical covariance for the uniform distributed rupture strength for different spatial resolutions ε.

particular for window sizes of $w = 1$ mm. The relative error between estimated and prescribed values is however independent of intensity or element size.

Turning to the case of fluctuating rupture strengths a similar picture emerges as shown in Fig. 8. In general, errors and standard deviations of the estimates are larger than for the uniform rupture strengths, in particular for the mean rupture strength f_0. Note that

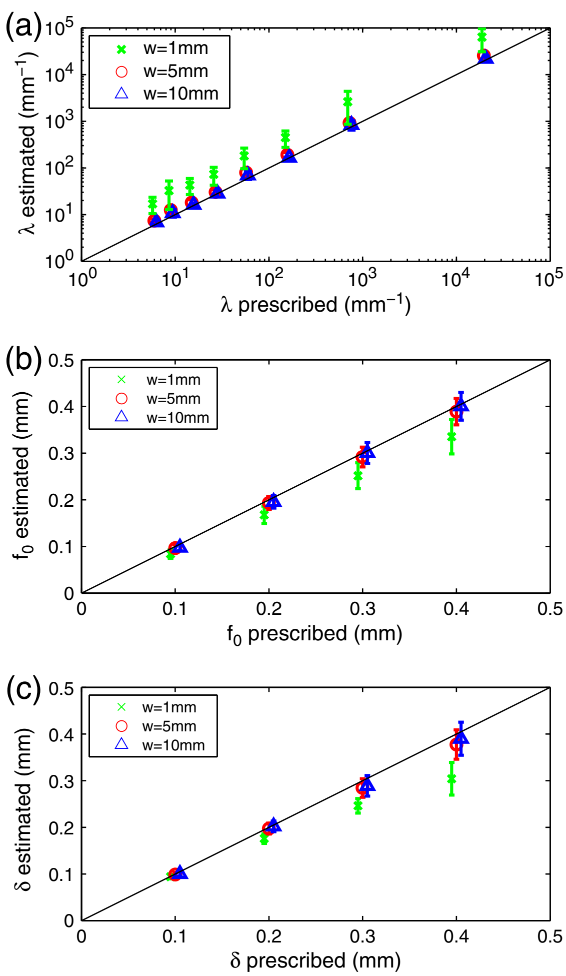

Fig. 7. Parameter estimation for idealized SMP signals with uniform rupture strengths for different window sizes w. (a) Intensity λ. (b) Rupture strength f_0. (c) Deflection at rupture δ.

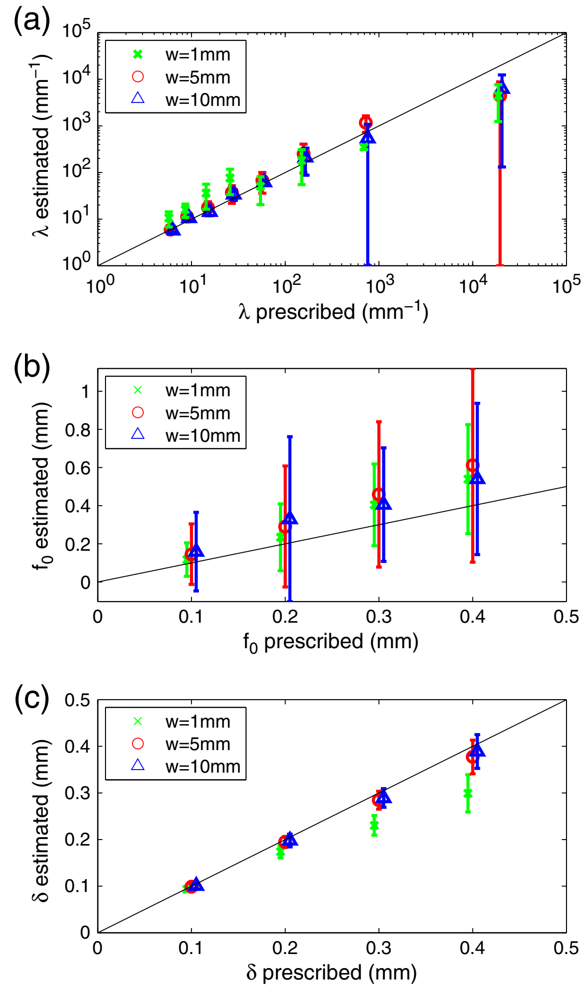

Fig. 8. Parameter estimation for idealized SMP signals with fluctuating rupture strengths for different window sizes w. (a) Intensity λ. (b) Rupture strength f_0. (c) Deflection at rupture δ.

the parameter α, as obtained from Eq. (14), often results in unphysical (negative) values. Thus, for a sharply peaked Gamma distribution ($\alpha = 10$), deviations from uniformity cannot always accurately be quantified. However, α estimates improve with increasing disorder, i.e. with decreasing α (not shown).

5. Application to a real snow profile

5.1. Parameter estimation

We apply our procedure to a natural snow profile observed at 2430 m a. s. l. on 31 January 2008 in the Eastern Swiss Alps near Davos, Switzerland. In addition, we evaluated the signal using the M&J method. This latter approach is based on peak counting and subsequent signal inversion. Peaks in the SMP data are attributed to the rupture of structural elements. To remove noise from the force signal, ruptures with a value less than a certain threshold based on the fluctuations of the SMP signal in air are removed. We used a default threshold value of 0.0165 N (Marshall and Johnson, 2009).

The SMP profile as well as the estimated parameters analyzed in 1 and 5 mm moving windows are shown in Fig. 9. The results show that increasing the window size essentially smooths the results. Based on manual snow cover observations (Fierz et al., 2009), the observed SMP force signal was divided into three portions: a thin surface layer consisting of new snow and two thicker layers consisting of

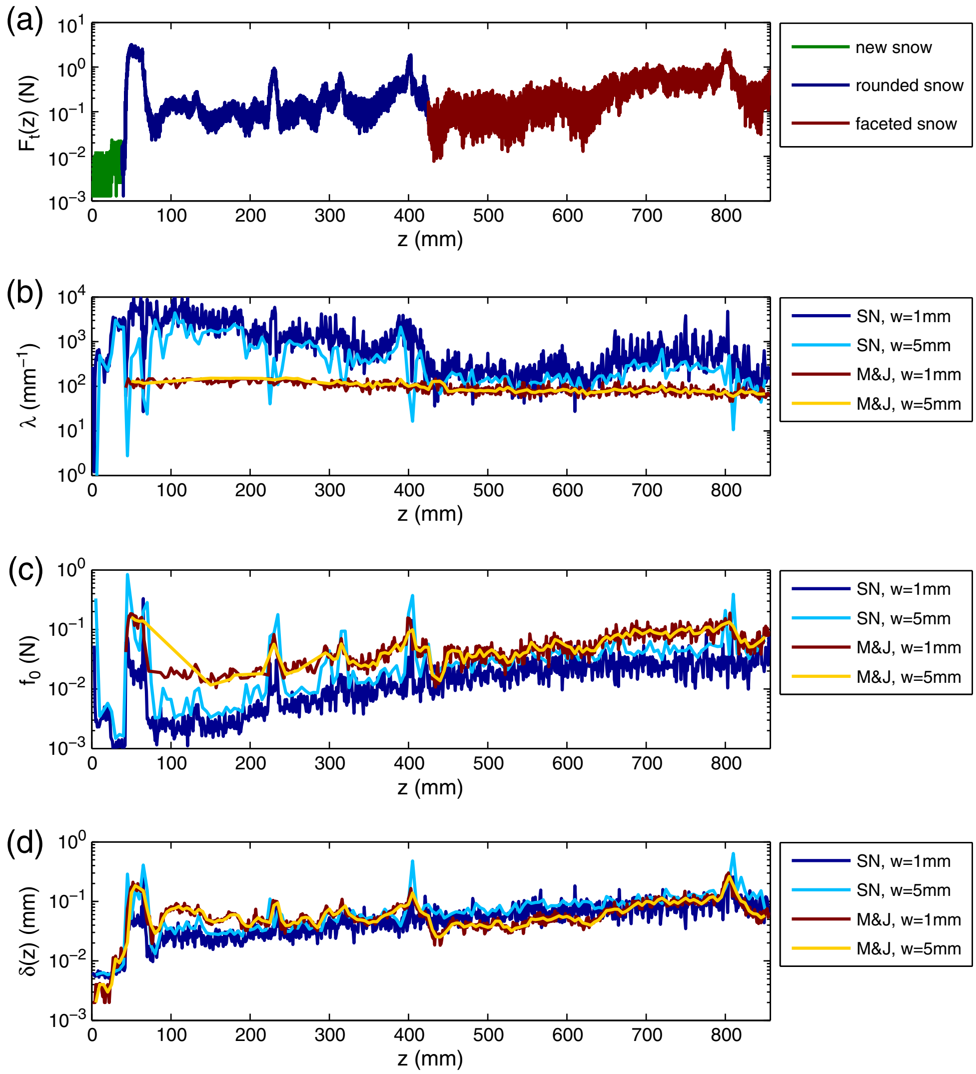

Fig. 9. Parameter estimation for different window sizes w according to (Marshall and Johnson, 2009) (M&J) and the prediction from the shot noise model (SN) for a real snow profile. (a) Force signals F_T from the original SMP profile. Three sections of the snow profile, mainly consisting of new snow, rounded snow and faceted snow, are highlighted. (b) Estimated intensity λ. (c) Estimated rupture strength f_0. (d) Estimated deflection at rupture δ.

mainly rounded and mainly faceted snow, respectively (Fig. 9a). The parameters estimated using the shot noise approach were different for each section of the signal. The intensity λ was largest for the rounded snow, while it was comparable for the new snow and the faceted snow (Fig. 9b). On the other hand, the rupture strength f_0 and the deflection at rupture δ generally increased with depth (Fig. 9c and d). Parameters estimated using the M&J method did not exhibit the same trends and were more uniform, particularly the intensity (Fig. 9b). Note that given the rupture threshold used to remove noise from the force signal, λ and f_0 could not be determined for the layer of new snow.

Comparison of the estimated parameters using the shot noise approach with those from the M&J method highlight the limitations of signal inversion by peak counting (Fig. 10). The derived element size using the M&J method was generally close to the lower limit below which signal inversion by peak counting is no longer possible, i.e. $L_{min} \approx 0.58$ mm. Using the M&J method the structural element size ranged from 0.5 to 0.75 mm, while a much broader range from 0.12 to 1.9 mm was estimated based on the shot noise approach (Fig. 10a). Since the derivation of f_0 in the M&J method is directly related to the derivation of λ, it comes as no surprise that the range of derived f_0 values was narrower than that obtained using the shot noise method (Fig. 10b). Finally, the deflection at rupture showed better agreement, as it is obtained independently from other parameters in the M&J method (Fig. 10c).

5.2. Comparison of force correlations

Finally, we briefly demonstrate that our formalism allows to validate the model assumptions which underlie the shot noise model and likewise the model from Marshall and Johnson (2009). A distinctive prediction of the present model is a correlation function which is, apart from its prefactor, completely characterized by the single length scale δ. This is an immediate consequence of both the absence of spatial correlations between element positions in the Poisson process and the single-scale force-displacement curve (1). We note that δ can be computed from Eq. (11) or, alternatively, from the first zero of the correlation function, cf. Eq. (10). The validity of the single-scale nature of $C(r)$ can thus be tested by comparing both methods for estimating δ for the real snow profile analyzed in the previous section. The results obtained for window size $w = 5$ mm are shown in Fig. 11a). Clearly, both methods result in different values of δ. We carried out the same analysis for a simulated shot noise signal (not shown) showing that both estimates generally coincide, though the

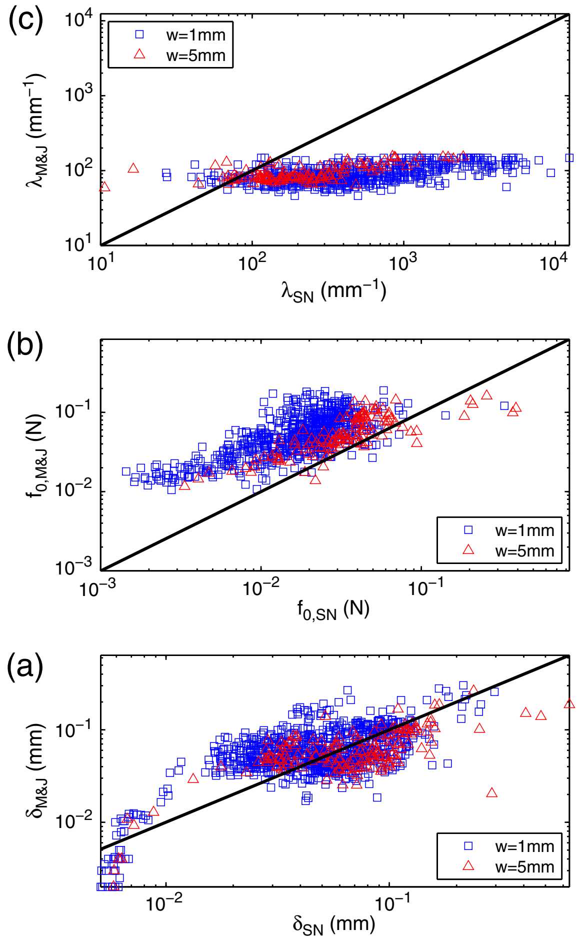

Fig. 10. Comparison of the estimated parameters for different window sizes w for the shot noise model (SN) and according to (Marshall and Johnson, 2009) (M&J) for the real snow profile shown in Fig. 9. (a) Estimated intensity λ. (b) Estimated rupture strength f_0. (c) Estimated deflection at rupture δ. The solid line indicates the 1:1 relation.

first zero of the correlation function is always subject to larger fluctuations. This finding suggests the existence of at least one additional relevant length scale in the force correlation function which is not predicted by Eq. (10). To corroborate this conclusion we have plotted the measured force correlation function at three different positions in the snowpack, as indicated by the arrows in Fig. 11a. These positions were subjectively chosen to include the following cases: (1) a large difference between both estimates for δ, (2) a position with a distinct peak in δ as obtained from Eq. (11), (3) a position where both estimates coincide. In addition we plotted the theoretically predicted $C(r)$ with δ from Eq. (11) as shown in Fig. 11b. This reveals that only at position 3 the functional form of $C(r)$ is well predicted by the spherical model. In contrast, at positions 1 and 2 force correlations persist over a larger spatial extent. A quantitative analysis of these tails of the correlation function is however beyond the scope of the present paper.

6. Discussion and conclusions

We have re-formulated the common micromechanical approach to cone penetration resistance in snow as a shot noise process which yields a simple, analytical method to estimate the micromechanical parameters from the statistics of the SMP force signal. The analytical solution reveals a direct link between the purely statistical approach (Satyawali and Schneebeli, 2010; Satyawali et al., 2009) and the micromechanical approach (Johnson and Schneebeli, 1999; Marshall and Johnson, 2009). From the analytical solution of the shot noise process with uniform rupture strengths we conclude that i) the normalized variance (the coefficient of variation) of the SMP force as used by Satyawali et al. (2009) equals 2/3 of the rupture strength f_0 (Eq. (12)) ii) the correlation length of the semi-variogram used by (Satyawali and Schneebeli, 2010) is given by δ (Eq. (10)) iii) the strong statistical dependence between the mean and the variance of the SMP force signal as found by Satyawali et al. (2009) can be explained by the dependence of these quantities on micromechanical parameters (Eq. (7)). Given the immediate implications, our stochastic micromechanical model opens the possibility of interpreting the snow classification (Satyawali and Schneebeli, 2010; Satyawali et al., 2009) in terms of mechanical parameters introduced in (Johnson and Schneebeli, 1999). It will be important to unify and systematically compare the present efforts with standardized snow samples to develop a common snow classification scheme. Our first, by no means exhaustive, comparison of the present model with the method from (Marshall and Johnson, 2009) has revealed similarities and also remarkable differences. A more detailed comparison is required and will be addressed in future work.

In the present work we have only given the simplest estimation procedure for the parameters in the case of uniform and fluctuating rupture strengths by employing the method of moments. Our goal was a simple analytical treatment of the parameter estimation problem to reveal the fundamental relations between micromechanical parameters and SMP statistics. We also demonstrated the limitations of the method of moments in the case of fluctuating rupture strengths. Though a robust estimate of the mean rupture force f_0 seems always feasible, a quantitative estimate of the degree of disorder via α is less reliable. A potential remedy might be an improvement of the parameter estimation procedure, e.g. by using a generalized method of moments or maximum likelihood estimators.

Our analytical solution of the stochastic model has revealed multiple means to validate the underlying model assumptions in measured SMP data. As indicated by our preliminary analysis of the force correlation function for a real snow profile, correlations are neither governed strictly by a spherical model nor by any other single-scale form such as an exponential, as suggested in (Satyawali and Schneebeli, 2010). We rather observe depth-dependent deviations of the inverse slope from the first zero of the (normalized) correlation function indicating at least two relevant length scales. This is in perfect agreement with the behavior of the density correlation function found in Löwe et al. (2011). The presence of multiple length scales is not very surprising since the snow microstructure must be regarded as a superposition of individual crystal scales and larger scale correlations which build up during aggregation (Löwe et al., 2007). Deviations of the experimental force correlation function from the model prediction (10) might originate from limitations of both, the Poisson point process for the distribution of elements or the ideal elastic–brittle mechanical behavior of an individual element. While sticking to the Poissonian assumption, which is actually more difficult to relax, deviations of the force correlation function from (10) have to be interpreted as an effective mechanical response which is *not* ideal elastic–brittle. A systematic comparison of SMP force correlations with density correlations obtained from micro-computed-tomography will be required to further elucidate these results. It will be helpful to extend the present analysis of force correlations in SMP signals also in view of generic properties of drag-force fluctuations in granular materials (Albert et al., 2000).

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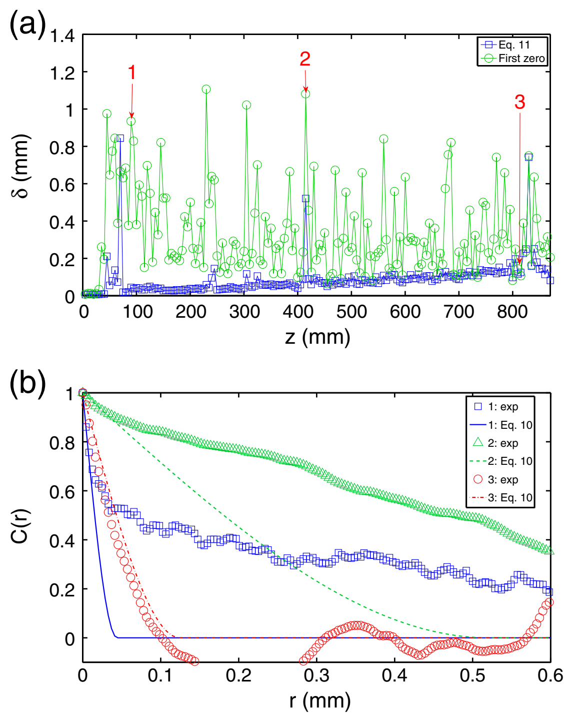

Fig. 11. (a) Comparison of different estimates of δ as a function of depth for the profile in Fig. 9. (b) Comparison of the experimental correlation function for the three positions from a) with the model prediction (10) where δ is obtained from Eq. (11).

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